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#SciTech കേരളം

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SCIENCE - TECHNOLOGY INNOVATION



അപ്പം തിന്നാൽ മതി, കുഴി ഞാൻ എണ്ണിക്കൊള്ളാം !

Just eat the appam; I will count the holes !



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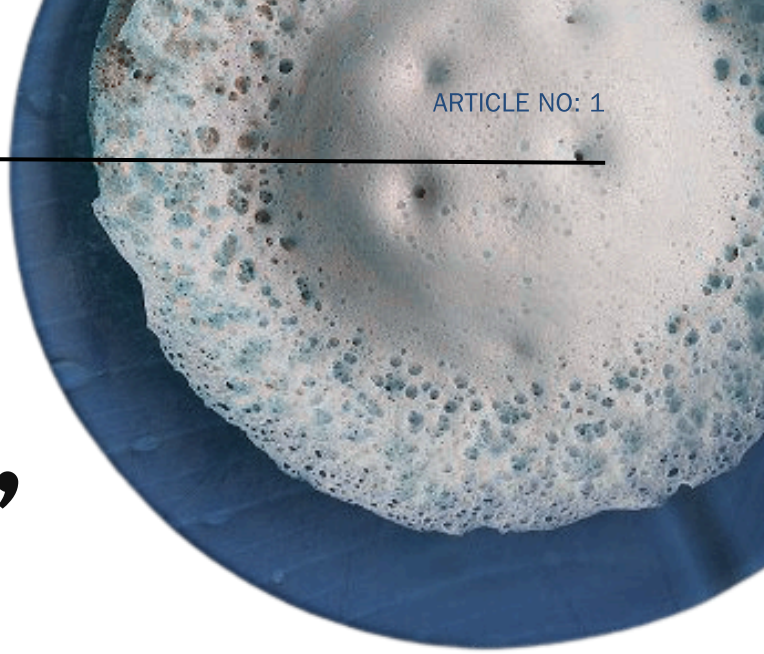


Prof Achuthsankar S Nair
Vice Chairman, Kerala State Higher Education Council

EDITORIAL NOTE

Science begins with a question, grows through curiosity, and becomes meaningful when it touches everyday life. This issue of #SciTechകേരളം brings together that spirit through ideas that are playful, practical, creative, and deeply human from counting the tiny holes in an appam to exploring AI, robotics, healthcare, education, and smarter ways of solving real-world challenges. What connects these stories is not just technology, but the imagination and courage of young minds willing to ask, “What if?” We hope these pages spark a question in you, inspire you to explore something new, and remind you that every great innovation can begin with a simple thought. To all our readers, may this issue kindle your curiosity, encourage you to experiment, and inspire you to turn your own ideas into possibilities. Happy reading and happy discovering!

SHIJIDA SHAIN
Chief Editor



APPAM

Thinnal Mathi, Kuzhi Njan Ennikolam !

In Kerala, there exists a popular Malayalam saying: *"Appam thinnal mathi, kuzhi ennenda,"* which roughly translates to *"Just eat the appam, don't count its holes."* The phrase is often used to tell someone not to overanalyze small details and to simply enjoy the outcome instead. For most people, the saying ends there. But for Aibel, it became a challenge.

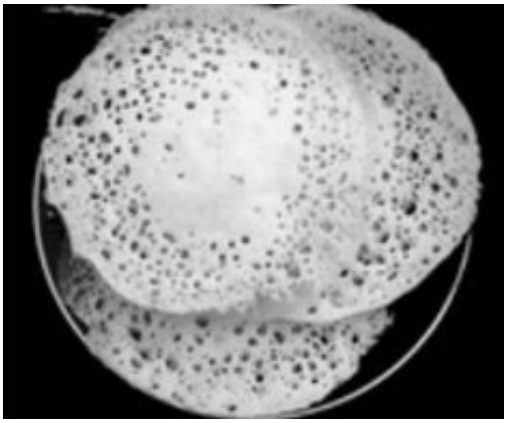


The idea for the project emerged during a casual classroom conversation after a teacher used the phrase in class. What started as a joke among classmates soon turned into a question that sounded nunnecessary yet strangely compelling: *What if someone actually counted the holes in an appam?*

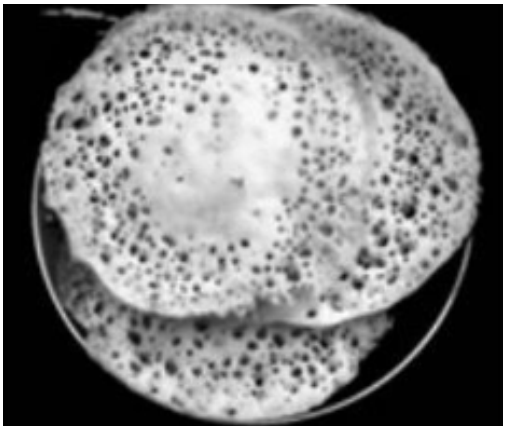
That single thought eventually led to the creation of **"Appam Thinnal Mathi, Kuzhi Njan Enniikolam"**, a program built specifically to detect and count the tiny holes found in an appam using image processing. The project was developed as part of the **"Useless Projects" Hackathon**, an initiative by TinkerHub that encouraged participants to build intentionally unconventional, humorous, and seemingly impractical ideas. This event celebrated curiosity, creativity, and the freedom to build things simply because they were interesting.

Developed with Python and lightweight image processing, the application by passed complex AI for a more elegant method. By overlaying a white intensity scale onto the appam image (inverse square law), it identified and counted darker regions representing holes using fundamental analysis principles. This simple, charming implementation stayed true to the project's spirit.

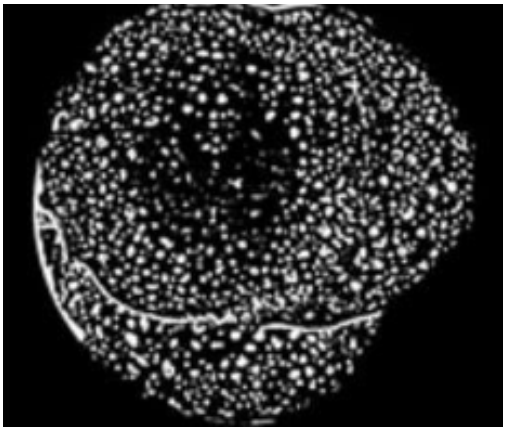
Aibel Bin Zacharia
Student - St. Joseph, Pala



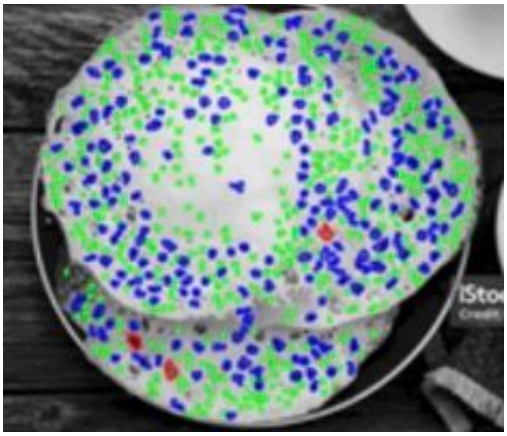
MASKED



ENHANCED



BINARY



FINAL DETECTION

The response was overwhelming. After being shared online, the project gained more than **200,000 impressions on LinkedIn**, reaching people far beyond the hackathon itself. Many appreciated the humor behind the concept, while others admired the creativity involved in taking such a random idea seriously enough to actually build it.

The project eventually became one of the winning entries of the hackathon, earning Aibel a six-month scholarship prize from TinkerHub. However, what made the experience truly impactful was not just the recognition, but the freedom that came with it.

For someone drawn toward unconventional ideas and experimental learning, this became an opportunity to dive deeper into technology in a personal and meaningful way. The scholarship support later helped fund future explorations, including brain-computer interface experiments, electronics purchases, cybersecurity tools, and access to AI development resources.

The success of the project stands as a celebration of playful engineering proof that technology can be meaningful not only when it solves problems, but also when it sparks joy.

DEMYSTIFYING PROCESSOR DESIGN

Vishwa R
Director, QinaPvt.Ltd.



Before we even dive into the design aspects, let's first understand what a processor really is. At its very core,a processor is nothing but a calculator. What does it calculate? It uses basic binary operations like AND, OR, NOT, XOR, and logical shifts to perform addition, subtraction, multiplication, and logical shifting of binary numbers. That's basically what a processor is. All of these are operations that aren't new to people;in fact, most of us already know how they work.

Now comes the real question: if this is all there is to a processor, then **why is it so hard to design one?**

The answer lies in how we actually feed data into it, how the data has to be processed and sent to the outside world, and the instructions used for these steps. This is no easy task to achieve. Other than just calculations, a processor is constantly moving data around, deciding when things should happen, and coordinating dozens of tiny operations with perfect timing.

In many ways, designing a processor is less like building a calculator and more like organising traffic in a massive city where millions of cars are moving at the same time without crashing into each

other. Achieving this feat is what makes processor design insanely complex. Modern processors are even capable of predicting outcomes, or more accurately, making educated guesses, and there is a whole separate hardware section dedicated to doing this.

Keeping all that aside, let's look at how one can go about designing a very minimal processor at home. Or is it even possible to do so? The answer is YES, we can. Surprisingly, the overall workflow is not very different from how professional companies do it. The scale is massively different, of course, but the core steps remain almost the same. A very basic processor can actually be designed by following these stages:

- Decide what the processor should do
- Design the Instruction Set Architecture (ISA)
- Break the processor into smaller parts. Create the digital logic
- Write the design in HDL
- Simulate the processor
- Implement the design on an FPGA
- Debug and improve the design
- Prepare the design for manufacturing
- Send the design to a fabrication company

What makes this fascinating is that almost every step up until manufacturing can genuinely be done from home with a computer, some free software tools, and an FPGA development board. The same fundamental ideas used by billion-dollar processor companies can be explored by hobbyists sitting at their desks. And even you.

And perhaps that is the most fascinating part of processor design. What powers our laptops, phones, and even massive supercomputers ultimately begins with a few simple logic operations and an idea sketched out by a human mind. Behind every modern computer lies an incredible blend of mathematics, engineering, creativity , and patience — all working together to turn tiny electrical signals into the digital world we interact with every single day.



TuTerra

A Student-Led EdTech Startup Redefining Modern Learning

പഠനം ഇനി വേറിട്ടൊരു അനുഭവം, വിദ്യാർത്ഥികൾ നയിക്കുന്ന **Tuterra**

TuTerra is emerging as a new-generation EdTech startup focused on transforming the way students experience tuition and academic support through technology-driven learning solutions. Incubated under Ideagora, St. Thomas College Ranni, TuTerra stands as an example of innovation led by college students who understand the real academic challenges faced by today's learners. The platform is built and managed by a passionate core team consisting of students from St. Thomas College, Ranni, reflecting the power of youth-driven entrepreneurship and educational innovation.

Midhun Mohanan
Operational Head,TuTerra



What makes TuTerra different is its student-centered vision and friendly learning culture. The platform focuses on providing supportive tutors who feel more like elder brothers and sisters rather than strict traditional teachers. This approach helps students communicate openly, learn comfortably, and build confidence in academics. It creates a positive learning environment where students can ask questions freely, understand concepts better, and receive personal guidance based on their needs. TuTerra provides both online and home tuition services, making quality education more flexible and accessible according to the needs of modern students and parents. With this blend of personal attention and flexible learning, TuTerra aims to make education a more comfortable and engaging experience.

One of the strongest pillars of TuTerra is its growing network of more than 700+ well-qualified professional tutors across multiple academic streams. From school-level subjects to higher secondary, degree-level, technical, and complex subjects including engineering streams, the platform connects students with experienced educators who focus on concept clarity, personalized mentoring, and effective teaching methodologies.

As a modern EdTech initiative, TuTerra strongly believes technology should play a major role in shaping the future of education. The platform actively trains tutors in modern teaching tools and innovative learning methodologies including Artificial Intelligence (AI), student-driven learning systems, digital teaching practices, modern educational technologies, and continuous upskilling techniques. Through technology-enabled communication, creative educational content, and innovative engagement strategies, TuTerra is building an ecosystem where learning becomes more interactive, motivating, and student-friendly.

TuTerra continues to grow as a youth-driven initiative committed to making meaningful change in the education sector. The initiative is guided by mentor Jikku James, along with the core team members Midhun Mohanan, Gouri Sudhir, Namitha Raj, Jyothish Jayan, Lekshmi S Nair, and Arsha Vinod, who collectively contribute towards building a modern educational ecosystem that combines quality mentorship, innovation, and technology for the next generation of learners.

TuTerra envisions an education system where technology and human mentorship work hand in hand. By making quality guidance more accessible and learning more personalized, the platform aims to empower students to learn with confidence, curiosity, and independence. Its growing community reflects a larger vision of education that is flexible, inclusive, and prepared for the future.

Beyond academics, TuTerra also aims to create a space where students feel motivated and supported throughout their learning journey. By combining personalized guidance with a friendly approach, the platform encourages students to become more confident, independent, and curious learners. It also helps them develop better learning habits and feel more comfortable asking questions. This focus on both academic growth and personal development makes the learning experience more meaningful and rewarding for every student in their everyday learning journey with confidence.

One One

CAN VOICE FEEL SOCIAL AGAIN?

There was a time when communication felt alive. You heard emotion in someone's voice before they even finished a sentence. Conversations were instant, natural, and uninterrupted. Today, despite having dozens of apps designed to "connect" us, most interactions have become slow, fragmented, and strangely distant. We type more than we talk. We wait for replies. We overthink messages. Even phone calls now feel formal, something that needs scheduling, commitment, or preparation.

That frustration is where One One began. I didn't want to build another social platform. I wanted to rebuild presence itself.

One One turns your phone into an instant voice channel. No dialling. No ringing. No waiting. You simply press and talk. The moment you press, your voice goes live. Release, and you're listening again. It's communication reduced to its purest form, immediate, real-time human interaction.

The idea came from observing how unnatural modern communication had become. Texting slows conversations down. Calls interrupt people's lives. Social media creates visibility, but not connection. Everything became optimised for content, not conversation.

Voice, however, is different.
The experience is intentionally simple. Push to talk. Always-on channels let people stay connected throughout the day without repeatedly starting or ending calls. Whether it's friends staying in sync, teams coordinating instantly, or people simply wanting to feel someone's presence in the background, One One makes communication feel lightweight and natural again.



Richard Shaju
Founder - OneOne, Mentor - Legacy IEDC UCEK

But building One One has not been easy. The biggest challenge is not technology, it's behavior. People have spent years adapting themselves to text-first communication. Convincing users to speak more naturally again requires more than features; it requires a cultural shift. Infrastructure for ultra-low-latency voice communication at scale is another challenge entirely. Real-time voice demands speed, reliability, and stability at every moment. Users expect conversations to feel effortless, and that invisible simplicity is incredibly difficult to engineer.

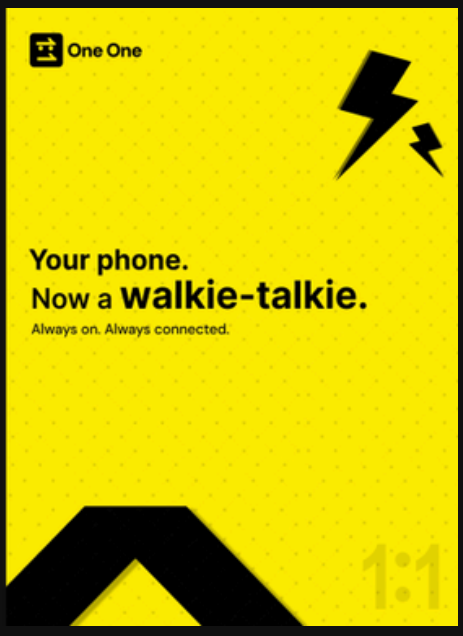
There is also the challenge of platform ecosystems themselves. Modern app stores and operating systems were not originally designed around persistent, live voice experiences.

Building something that feels truly instant while navigating technical and platform limitations has been one of the hardest parts of the journey. Modern app stores and operating systems were not originally designed around persistent, live voice experiences. Ensuring smooth communication while maintaining reliability, speed, and simplicity requires constant testing and refinement. Still, the vision remains clear: to build an experience that feels natural, responsive, and effortless for everyone.

I believe communication in the future will move away from polished feeds and back toward real presence. We are entering a time where people are exhausted by performing online. They want interactions that feel immediate, authentic, and alive again. One One is not trying to replace messaging apps or social networks. It is trying to create a new layer of communication, one where talking becomes as frictionless as thinking.

The long-term vision is ambitious: a world where voice becomes ambient, instant, and always available. A world where distance disappears not because of better content, but because of better presence.At its core, One One is built on a simple belief: Sometimes the fastest way to feel close to someone is just hearing their voice. In the future, the platform aims to make these moments feel even more natural, seamless, and accessible, allowing people to stay connected without the barriers of traditional communication.





THE POCKET-SIZED DOCTOR

that sees what we miss

A new framework runs entirely on a device smaller than a paperback—and it could save millions from preventable blindness.

Somewhere in a rural clinic, with no reliable broadband and a single overworked nurse, a patient with diabetes sits waiting. Their eyes look fine. They feel fine. But inside the retinal that postage-stamp of light-sensitive tissue at the back of the eye blood vessels are quietly failing. Without a specialist to look, the damage will go unnoticed until it is too late.



This is the hidden crisis of diabetic retinopathy: a condition that affects nearly one in three people with diabetes and is the leading cause of preventable blindness in working-age adults worldwide. The tragedy is that early detection works. Treatment, started in time, can halt progression almost entirely.

"The bottleneck is not medicine — it is access. A new generation of artificial intelligence is now small enough to go wherever patients are." It was found that the AI-powered screening system could run entirely on the NVIDIA Jetson Nano — a computing board roughly the size of a credit card — while consuming less power than a desk lamp. It processes retinal photographs, classifies disease severity across five clinical grades, and explains its reasoning to the clinician, all without sending a single pixel to the cloud.

The system uses two versions of a neural architecture, which was designed from the ground up to do more with less. The lighter variant serves as a rapid first-pass filter. The heavier one delivers the detailed verdict.

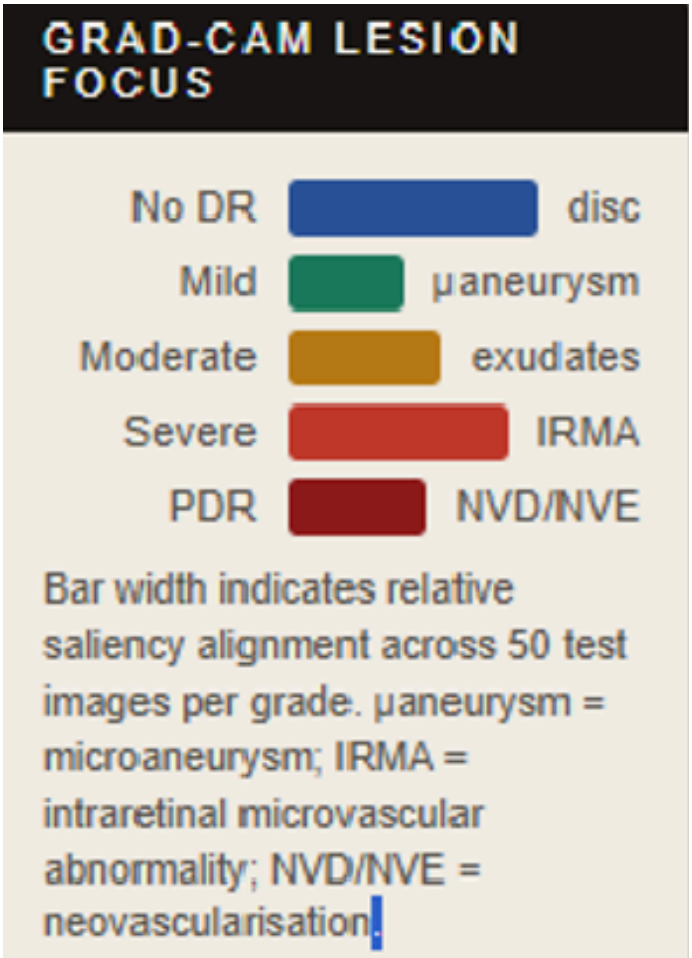
Before either model sees an image, the photograph passes through a five-stage conditioning pipeline: borders stripped, lighting corrected, the green colour channel — where blood vessel contrast is sharpest , selectively sharpened using

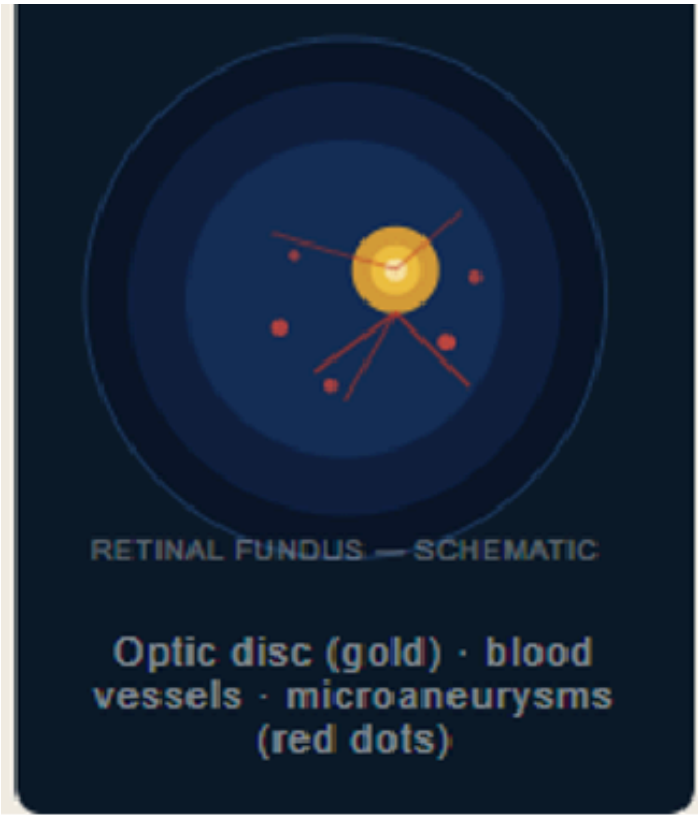
Shijida Shain
Asst. Prof., University College of Engineering, Kariavattom

a technique called CLAHE. Finally, the image is normalised to ensure consistent intensity and colour distributions across different cameras and acquisition conditions. The result is a cleaner, standardised retinal image that allows the neural networks to focus on clinically relevant features rather than variations introduced during imaging.

Equally important is the system's ability to explain its decisions. Clinicians are understandably hesitant to trust a model that simply outputs a disease label without justification. To improve transparency and support clinical decision-making, the framework incorporates Gradient-weighted Class Activation Mapping (Grad-CAM), generating heat maps that highlight the retinal regions most responsible for each prediction. Areas containing microaneurysms, retinal haemorrhages, hard exudates, cotton-wool spots, or abnormal neovascularisation are visually emphasised and overlaid directly on the original fundus photograph. These explanations are generated in real time on the same Jetson Nano device, allowing healthcare professionals to verify the AI's reasoning rather than treating it as a black box. By combining efficient edge computing with explainable artificial intelligence, the system offers a practical and trustworthy solution for early diabetic retinopathy screening in underserved communities.

By combining efficient deep learning, and edge computing, this framework demonstrates that accurate diabetic retinopathy screening is no longer confined to specialised hospitals. It brings real-time diagnosis closer to underserved communities, making early detection more accessible and reducing the risk of preventable vision loss through timely clinical intervention.





BALANCING THE SCALES

The Student-Led Innovation Fighting Unfair Insurance Denials

ഇൻഷുറൻസ് നിഷേധിച്ചാൽ, ഇനി ചോദിക്കാൻ AI ഉണ്ടാകും

A Providence College of Engineering student team uses new technology to protect families from unexpected financial stress during health emergencies.

The core of this project is a painful reality observed in hospitals today: families face emotional trauma during a medical emergency, only to receive a sudden notice that their health insurance claim has been denied. For most middle and lower-class families, health insurance should provide essential financial protection. Instead, many valid claims get denied due to confusing paperwork, strict policy rules, or minor documentation errors. Watching people face severe debt or struggle to get necessary medication because they couldn't understand complicated insurance terms inspired a group of B.Tech Artificial Intelligence students at Providence College of Engineering. Guided by their mentors, Ms Jini George and Mr Prem Sankar C, they aimed to create a digital advocate for everyday people.

Driven by a wish to help, the student team—Harini M. Anand, Fathima Neha T.P., Nevin Mathew Benny, and Sarthik S. Kumar—developed ClaimAssist AI. The software functions as an automated, intelligent consumer advocate. It securely reviews complex medical records and the user's insurance policy, automatically highlights unfair discrepancies, and assesses whether a denial is legally valid. Most importantly, if the denial is deemed unfair, the system automatically generates a professional,



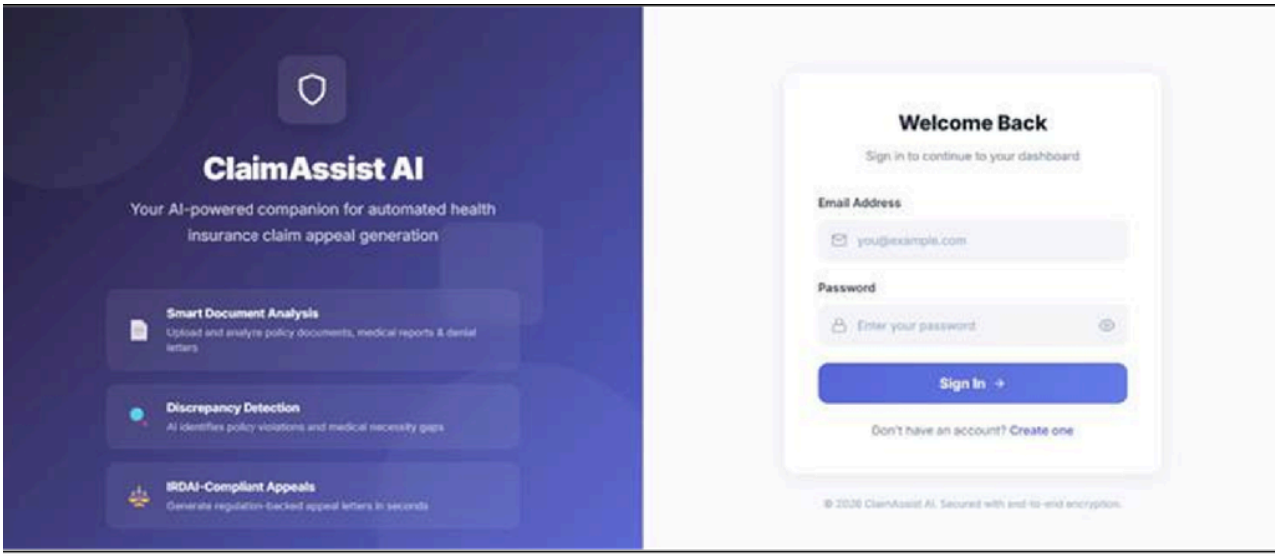
evidence-based appeal letter designed to withstand unjust corporate rejections. An official appeal letter is a formal request asking an insurance company to reconsider its denial.

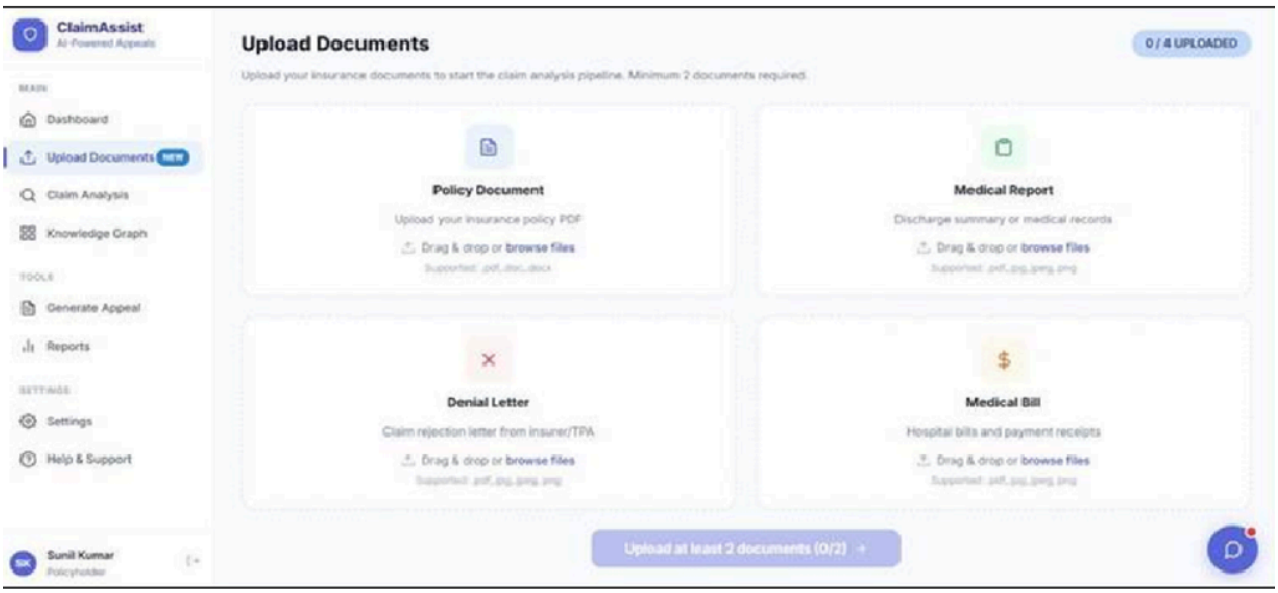
"ClaimAssist AI transforms scared patients into informed, confident advocates, ensuring that financial safety nets actually work when families need them most."

What makes this project a significant breakthrough is its technical reliability coupled with speed. While typical AI models may create inaccuracies, this framework achieves expert-level accuracy by using a decentralised multi-agent system that processed 100

complex legal and medical documents in under 40 seconds during testing. By dividing the analytical tasks among specialised virtual "agents"—one checking insurance rules, another reading clinical codes, and another examining low-quality hospital notices —it replaces costly legal fees and weeks of manual document searches with immediate, clear results. Choosing ClaimAssist AI removes traditional obstacles that leave policyholders confused, forced to interpret dense legal terms alone while managing a healthcare crisis.

The broader mission of ClaimAssist AI is to become a vital safety net for Indian society. It targets the root causes of medical bankruptcy among middle and lower-class families by making the appeal process completely accessible and cost-free. Looking ahead, the team plans to expand its local prototype into a scalable enterprise solution, to support multiple users at once while keeping operational delays low. Ultimately, ClaimAssist AI stands out as a scalable social enterprise designed to reduce financial stress, and empower millions of patients across the country.





PREDICTIVE WEATHER ROUTE PLANNER

വഴിയിൽ മഴയുണ്ടോ? യാത്ര തുടങ്ങും മുമ്പേ അറിയാം!

The idea for the Predictive Weather Route Planner was born during our weekend two-wheeler trips to Ponmudi. As riders, we were fully exposed to the weather, often caught off guard by sudden downpours or harsh, exhausting sunlight. One afternoon, while riding through the hairpin curves of Ponmudi, we encountered a severe and unexpected rainstorm. Although the city forecast showed clear weather, the mountain roads had become slippery and unsafe. That experience made us realise the need for weather predictions that are not just location-based, but route-specific and time- sensitive.

Our primary objective is to predict weather conditions along the actual route, at the exact time a rider is expected to pass through each point. By calculating estimated arrival times at the starting point, midpoint, and destination, and matching them with future three-hour forecast intervals, the system provides timely safety alerts before the journey begins.

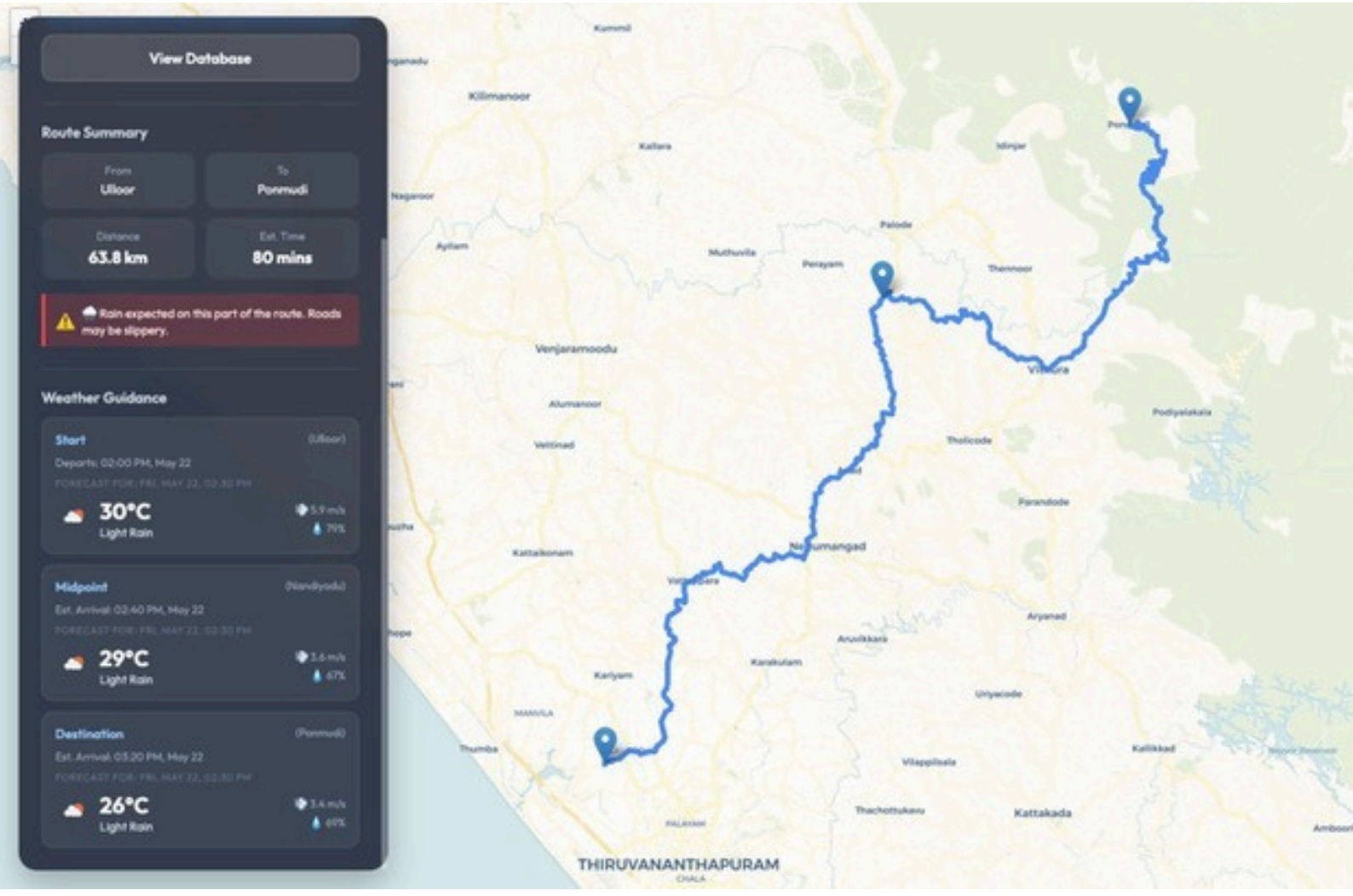
Nandhakrishna VS , Riya Roy, Pavithra R Raj, Nimish M Biju, Nihal Mohhamed Shafy
University College of Engineering, Kariavattom



Building the application came with its own challenges. Initially, we found that there were no simple libraries that could directly connect Leaflet Routing Machine coordinates with historical or forecast weather data. To overcome this, we developed custom JavaScript functions to calculate road midpoints and used JavaScript Sets to filter duplicate weather alerts efficiently.

This project was a collaborative effort by our team as part of our third-year mini project. Under the guidance of Ms. Shijida Shain, the project helped us understand how to connect interactive map interfaces with secure backend systems.

At present, we are refining the initial prototype by improving geocoding debounce rates and caching forecast tokens. Looking ahead, we hope to develop this prototype further as our fourth-year final project, with future plans including migration to a robust cloud database infrastructure, integration of real-time traffic density mapping, and incorporation of machine learning models for high-resolution microclimate predictions along mountain routes.



CrowdVote AI

തിരഞ്ഞെടുപ്പ് പ്രവചനത്തിന്റെ പുതിയ വഴി

CrowdVote AI is a real-time electoral forecasting platform developed for the 2026 Kerala Assembly Elections. The project is based on the idea of Swarm Intelligence, where the collective choices and predictions of a large number of people are used to identify possible election trends and outcomes.

I developed this project as an experiment in combining technology, public participation, and data- driven prediction systems. The project was developed by Aaron Alex Mathew, under the guidance of Dr. Prem Sankar Sir, Head of the Department, Providence College of Engineering. In CrowdVote AI, users can select constituencies and predict which political front is most likely to win. Instead of simple polling, the platform assigns a score called Signal Strength to each prediction. The score depends on two factors: the confidence level of the user and the timing of the prediction during different election phases. Predictions made with higher confidence receive greater weight in the final analysis.



The platform collects these predictions from thousands of users and combines them into a live forecasting system using swarm intelligence — the idea that the combined judgment of a large group can often produce meaningful insights. Users can also view leaderboards, live analytics, and constituency-wise trends, making the experience interactive and engaging. The system was built using React, Node.js, Express, and Firebase, with a secure backend for handling predictions and real time data aggregation.

CrowdVote AI demonstrates how modern web technologies and collective human intelligence can be used to create a dynamic and participatory approach to electoral analysis.

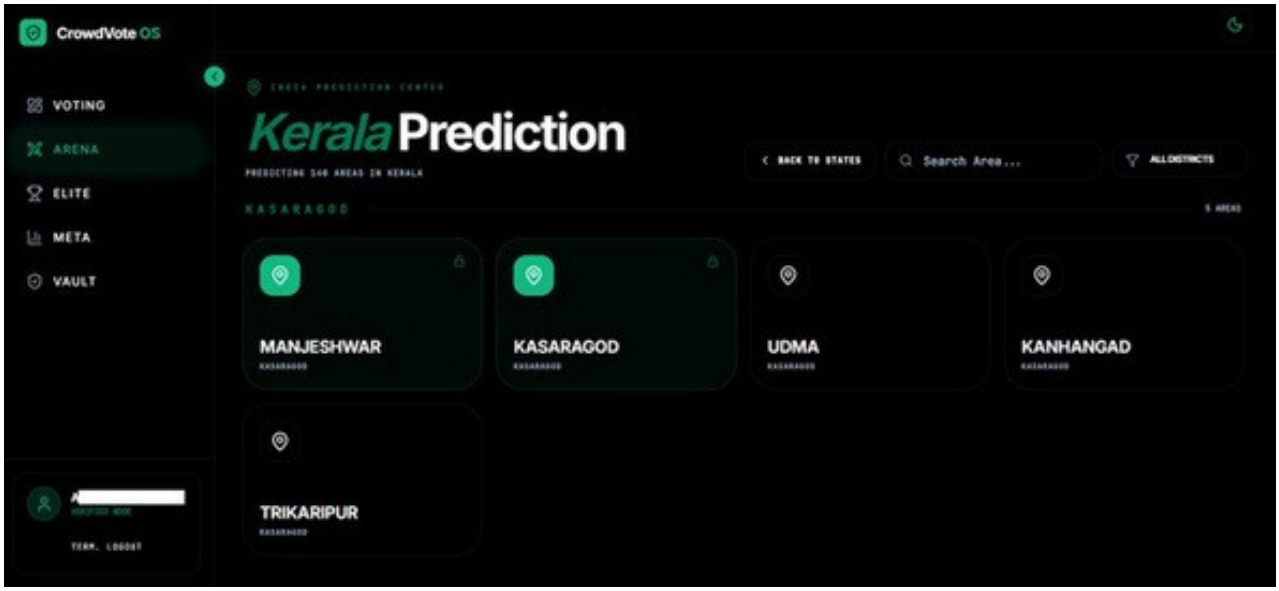
Aaron Alex Mathew
Providence College of Engineering



Home Page



Admin Page



Voting Page

Generative AI-Based Personalized Movie Explanation System

ഈ സിനിമ എന്തുകൊണ്ട് കാണണം? ഇനി AI പറഞ്ഞുതരും!



Julienne Brijit Jacob
Student, Providence College of Engineering

With hundreds of movies releasing across theatres and streaming platforms every month, choosing what to watch has become surprisingly difficult. Most recommendation systems suggest movies based on ratings, genres, or viewing history, but they rarely explain why a particular movie matches a user's interests. Recognizing this gap, we (**Adithyan Sumesh, Ashik Ajil, and Julienne Brijit Jacob**), the students from the Department of Computer Science and Engineering (Artificial Intelligence), Providence College of Engineering, Chengannur, under the guidance of Dr. Prem Sankar C, developed the Generative AI-Based Personalized Movie Explanation System — an AI-powered platform designed to provide personalized movie insights.

Unlike traditional recommendation systems that function like "black boxes," our platform attempts to interact with users in a more human-like way. Instead of simply recommending a movie, the system explains why the movie may suit a user's preferences, mood, favourite genres, or viewing interests. We aimed to make movie discovery more meaningful and trustworthy by combining modern Generative AI technologies with explainable reasoning.

The project uses Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and Agentic AI workflows to generate these personalized explanations. In simple terms, the system first searches for relevant movie information from its knowledge base and then interfaces with the advanced OpenAI Application Programming Interface to create a natural-language explanation tailored to the user's profile. This approach eliminates the common risk of "made-up" or hallucinated AI responses, ensuring that every explanation remains strictly grounded in verified movie data.

To build the system, we compiled information on nearly 2000 movies using sources such as the TMDB API and Wikipedia. The application uses semantic search technology to understand the meaning behind user queries instead of relying only on exact keywords. For example, if a user searches for “emotional sci-fi movies with strong character development,” the system can identify movies with similar themes even if those exact words are not present in the description.

We developed the platform as a full-stack web application featuring a highly responsive React-based user interface, a Node.js backend, and a dedicated PythonAI service. By physically separating the standard web layer from the heavy AI processing pipelines, our team ensured a smart engineering design: the platform remains incredibly fast and maintains low-latency performance even under high traffic. The project also includes Text-to-Speech (TTS) functionality, allowing these conversational explanations to be converted into accessible audio streams for improved user engagement.



KEY HIGHLIGHTS

- Personalized Reasoning: Uses Generative AI to provide descriptive movie explanations instead of static percentage matches or generic recommendations.
- Contextual Accuracy: Combines advanced RAG frameworks and semantic search to understand the true intent behind user queries.
- Zero Hallucinations: Eliminates incorrect AI- generated responses by strictly restricting the AI's generation to verified movie data vectors.
- Multimedia Accessibility: Supports natural- language interaction and integrated Text-to- Speech audio explanations.
- High-Performance Engineering: Developed as a scalable microservices web application that splits heavy AI workloads away from the core web server to guarantee sub-millisecond retrieval speeds.

Testing showed that the RAG-and-agent framework successfully produced highly relevant, context-aware explanations that far surpass conventional chatbot paradigms. The project demonstrates how Generative AI can move beyond standard chatbots to create more transparent, interactive, and user-friendly entertainment systems. As AI continues to become a part of everyday life, projects like this highlight how young researchers are exploring innovative ways to combine technology, personalization, and explainability to improve user experiences in real-world applications.

DESKTOP AI ROBOT

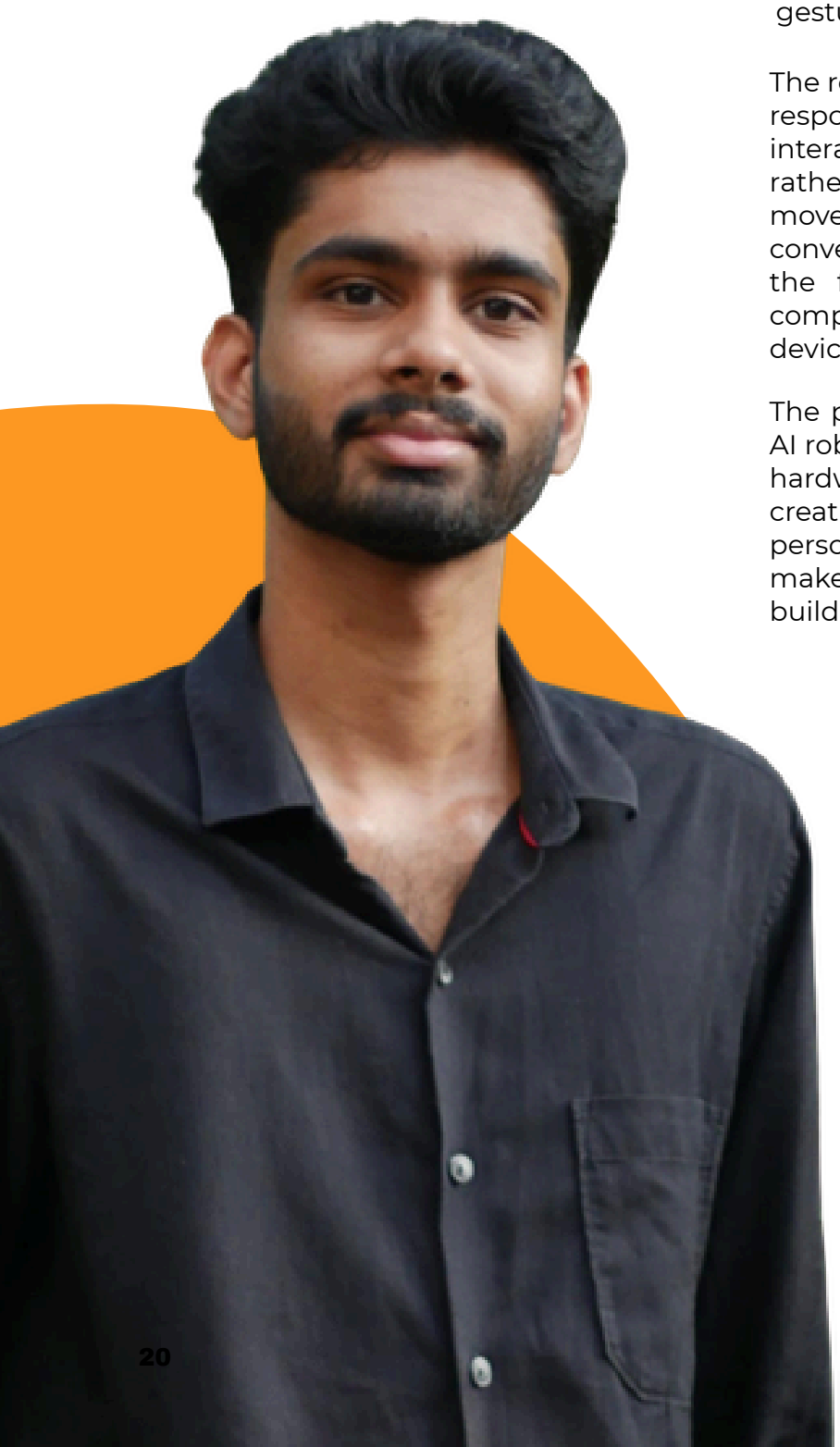
A Smartphone- Powered Robotic Companion

The Desktop AI Robot is a compact, affordable robotic companion designed to sit on a desk and provide genuine AI interaction. The project is built around one simple idea: use existing technology efficiently instead of relying on expensive custom hardware. Instead of embedding a dedicated display or AI processor, the robot docks a smartphone, inspired by the Looi Robot. The smartphone acts as the robot's face and intelligence system, handling voice recognition, camera input, expressions, and AI processing, while the robot body provides movement and physical interaction.

The hardware is built around an ESP32 microcontroller that manages servo movement and wireless communication. It receives commands from the smartphone, controls the motors, and keeps the system responsive in real time. The smartphone handles AI processing, while the ESP32 focuses on physical control and movement.



For on-device AI, the project uses Llama 3.2 1B together with MediaPipe, both running directly on the smartphone. This allows the robot to process voice input, recognize gestures, and respond naturally without requiring a constant internet connection. Running AI locally keeps interactions fast, private, and self-contained. When internet access is available, the robot can connect to larger AI models for advanced reasoning and demanding tasks.



WHAT'S INSIDE

ESP32 for movement control and wireless communication.

Smartphone for voice assistant, camera input, display, and AI processing.

Llama 3.2 1B for lightweight on-device language processing.

MediaPipe for real-time vision and gesture recognition.

The robot is designed to move naturally, respond to voice in real time, and interact in a way that feels expressive rather than mechanical. Through movement, expressions, and conversational AI, the system creates the feeling of a responsive desktop companion instead of a static electronic device.

The project demonstrates that capable AI robotics can be built using accessible hardware and modern AI software, creating a practical approach to personal robotics that students, makers, and developers can realistically build and improve.

ADWAYTH LAL S
Quality And Operations Officer
Legacy IEDC UCEK

ഡാറ്റാനുഭവം !!! ആഹാ

“AI എത്ര ബുദ്ധിയുള്ളതായാലും മനുഷ്യന്റെ സൃഷ്ടിപരതയെ മറികടക്കാൻ കഴിയില്ല.” ഇന്നും കലാകാരന്മാരുടെയും സാഹിത്യകാരന്മാരുടെയും വായിൽ ഏറ്റവും കൂടുതലായി കേൾക്കുന്ന വാചകങ്ങളിൽ ഒന്നാണിത്. പക്ഷേ, ഈ അഭിപ്രായം പലപ്പോഴും ആഴത്തിലുള്ള പഠനത്തിൽ നിന്നല്ല, മറിച്ച് ഒരു മുൻവിധിയിൽ നിന്നോ സാങ്കേതികവിദ്യയെ അടുത്തറിയാത്തതിൽ നിന്നോ വരുന്നതാകാമോ?

“പലരും ഇപ്പോഴും AI-യെ “ചോദ്യങ്ങൾക്ക് ഉത്തരം പറയുന്ന ഗൂഗിൾ” എന്ന നിലയിലാണ് കാണുന്നത്. എന്നാൽ ഇന്നത്തെ Large Language Models (LLMs) — ChatGPT പോലുള്ള സംവിധാനങ്ങൾ — വെറും വിവരശേഖരങ്ങൾ അല്ല. അവ കഥ പറയുന്നവരാണ്. കവിത എഴുതുന്നവരാണ്. തമാശ പറയുന്നവരാണ്. ചിലപ്പോൾ നമ്മെ തന്നെ ആശ്ചര്യപ്പെടുത്തുന്ന രീതിയിൽ മനുഷ്യഭാവങ്ങളെ അനുകരിക്കുന്നവരുമാണ്.

മലയാള കവിയായ മധുസൂദനൻ സാറിന്റെ പ്രതികരണം ശ്രദ്ധേയമാണ്. AI എഴുതിയ ഒരു കവിത അദ്ദേഹത്തെ കാണിച്ചപ്പോൾ, പലരും പ്രതീക്ഷിച്ചതുപോലെ “ഇത് യന്ത്രത്തിന്റെ പകർത്തൽ” എന്ന് അദ്ദേഹം പറഞ്ഞില്ല. പകരം അദ്ദേഹം എഴുന്നേറ്റ് സ്ക്രീനിനെ സല്യൂട്ട് ചെയ്തു. അദ്ദേഹം പറഞ്ഞത് ഇതായിരുന്നു: “AI-യ്ക്ക് ഇത്രയും സൃഷ്ടിപരതയുണ്ടെങ്കിൽ അതിനെ ആദരിക്കാൻ ഞാൻ തയ്യാറാണ്.”

LLM എന്നാൽ എന്ത്?

LLM എന്നത് Large Language Model എന്നതിന്റെ ചുരുക്കപ്പേരാണ്. കോടിക്കണക്കിന് പുസ്തകങ്ങളും ലേഖനങ്ങളും സംഭാഷണങ്ങളും പഠിച്ച ശേഷം ഭാഷയിലെ പാറ്റേണുകൾ മനസ്സിലാക്കുന്ന ഒരു സംവിധാനമാണത്. പക്ഷേ ഒരു സംശയം ഉടനെ വരും:“പഠിച്ചതെല്ലാം ചേർത്ത് തിരികെ പറയുന്നതല്ലേ ഇത്?” ഭാഗികമായി ശരിയാണ്. പക്ഷേ മനുഷ്യന്റെ തലച്ചോറും അതേ കാര്യമാണ് ചെയ്യുന്നത്. ഒരു കവി തന്റെ ജീവിതത്തിൽ കേട്ട ഭാഷ, വായിച്ച പുസ്തകങ്ങൾ, കടന്നു പോയ പ്രണയങ്ങൾ, അനുഭവിച്ച ദുഃഖങ്ങൾ —



Aparna Sudhakaran
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“AI എത്ര ബുദ്ധിയുള്ളതായാലും മനുഷ്യന്റെ സൃഷ്ടിപരതയെ മറികടക്കാൻ കഴിയില്ല.” ഇന്നും കലാകാരന്മാരുടെയും സാഹിത്യകാരന്മാരുടെയും വായിൽ ഏറ്റവും കൂടുതലായി കേൾക്കുന്ന വാചകങ്ങളിൽ ഒന്നാണിത്. പക്ഷേ, ഈ അഭിപ്രായം പലപ്പോഴും ആഴത്തിലുള്ള പഠനത്തിൽ നിന്നല്ല, മറിച്ച് ഒരു മുൻവിധിയിൽ നിന്നോ സാങ്കേതികവിദ്യയെ അടുത്തറിയാത്തതിൽ നിന്നോ വരുന്നതാകാമോ?

സൃഷ്ടിപരത എന്നത് എന്താണ്?

നമ്മൾ പലപ്പോഴും creativity എന്നത് “ആകാശത്തു നിന്നു പൊട്ടിവീണദിവ്യപ്രകാശം” പോലെ ആണെന്ന് ചിന്തിക്കാറുണ്ട്. എന്നാൽ cognitive science പറയുന്നത് മറ്റൊന്നാണ്. സൃഷ്ടിപരത പലപ്പോഴും പഴയ ആശയങ്ങളുടെ പുതിയ കൂട്ടിച്ചേർക്കൽ, അപ്രതീക്ഷിത ബന്ധങ്ങൾ കണ്ടെത്തൽ, ഭാഷയിലെയും ചിന്തയിലെയും playfulness, pattern breaking, emotional resonance സൃഷ്ടിക്കൽ എന്നിവയുടെ സമന്വയമാണ്. ഇതിൽ പലതും ഇന്നത്തെ AI സംവിധാനങ്ങൾ അതിശയകരമായി ചെയ്യുന്നു. AI-യോട്: “കുഞ്ഞൻ നമ്പ്യാരുടെ ശൈലിയിൽ ഒരു രാഷ്ട്രീയ തുള്ളൽ എഴുതൂ”, “രാത്രിമഴയെക്കുറിച്ച് സച്ചിദാനന്ദൻ ശൈലിയിൽ കവിത രചിക്കൂ”, “ഒരു മീമിന്റെ ഭാഷയിൽ quantum physics വിശദീകരിക്കൂ” എന്നൊക്കെ പറഞ്ഞാൽ അതു പലപ്പോഴും നമ്മെ ചിരിപ്പിക്കുകയും വിസ്മയിപ്പിക്കുകയും ചെയ്യുന്ന output നൽകും.

എന്നാൽ മനുഷ്യ കലാകാരൻ അപകടത്തിലാണോ?

ക്യാമറ വന്നപ്പോൾ ചിത്രകാരന്മാർ ഭയപ്പെട്ടു. സിന്തസൈസർ വന്നപ്പോൾ സംഗീതജ്ഞർ ഭയപ്പെട്ടു. Photoshop വന്നപ്പോൾ ഡിസൈനർമാർ ഭയപ്പെട്ടു. ഇപ്പോൾ AI വന്നപ്പോൾ എഴുത്തുകാരും കലാകാരന്മാരും ഭയക്കുന്നു. പക്ഷേ ചരിത്രം പറയുന്നത് ഇതാണ്: പുതിയ സാങ്കേതികവിദ്യ പഴയ കലകളെ ഇല്ലാതാക്കാറില്ല; മറിച്ച് അവയെ മാറ്റിമറിക്കാറാണ് പതിവ്. ഇന്നത്തെ കലാകാരൻ ഒരേസമയം: creator,curator,prompt engineer,editor, performer എന്നിവയൊക്കെയായി മാറുകയാണ്. AI ഒരു “replacement” മാത്രമല്ല; അത് ഒരു “creative amplifier” കൂടിയാണ്.

പക്ഷേ ജാഗ്രതയും വേണം

AI: fake quotes ഉണ്ടാക്കാം, misinformation സൃഷ്ടിക്കാം, stereotypes reinforce ചെയ്യാം, plagiarism പോലുള്ള പ്രശ്നങ്ങൾ ഉണ്ടാക്കാം. അതുകൊണ്ട് critical thinking അനിവാര്യമാണ്. AI ഉപയോഗിക്കുന്ന തലമുറയ്ക്ക് creativity-ക്കൊപ്പം scientific temper-വും പ്രധാനമാണ്. മനുഷ്യൻ സൃഷ്ടിച്ച യന്ത്രം ഇപ്പോൾ മനുഷ്യനെപ്പോലെ കവിത എഴുതുന്നു. ഇത് ഭയപ്പെടേണ്ട കാര്യമോ? അതോ ആഘോഷിക്കേണ്ട കാര്യമോ? രണ്ടും. എന്നാൽ ഒരു കാര്യം ഉറപ്പാണ്: മനുഷ്യന്റെ സൃഷ്ടിപരതയെ കുറിച്ച് നമ്മൾ കരുതിയിരുന്ന പല “പവിത്ര മിത്തുകളെയും” AI ചോദ്യം ചെയ്തുകഴിഞ്ഞു. മധുസൂദനൻ സാർ സ്ക്രീനിനെ സല്യൂട്ട് ചെയ്തത് അതുകൊണ്ടാവാം. അദ്ദേഹം ഒരു യന്ത്രത്തെ ആദരിച്ചില്ല. മറിച്ച്, മനുഷ്യബുദ്ധി സൃഷ്ടിച്ച പുതിയൊരു ബുദ്ധിയുടെ ഉദയത്തെ അംഗീകരിക്കുകയായിരുന്നു. ഭാവിയിലെ ഏറ്റവും വലിയ കലാകാരൻ “മനുഷ്യൻ” ആകുമോ? അതോ “AI” ആകുമോ? രസകരമായ ഒരു സാധ്യതയുണ്ട്: മനുഷ്യനും AI-യും ചേർന്നുള്ള ഒരു പുതിയ 'ജീവി' ആയിരിക്കാം അത്.



Tracking Life Beyond Glucose:

The Power of CGM in Type 1 Diabetes Care

At the age of 12, my life took an unexpected turn when I was diagnosed with Type 1 Diabetes. Now, at 19, I have spent eight years learning not just how to manage it, but how to live alongside it. Type 1 Diabetes is a lifelong condition that demands constant attention, discipline, and resilience. Every day involves careful decisions about food, insulin, activity, and rest, shaping me into someone more aware, responsible, and strong.

Type 1 Diabetes is an autoimmune disease in which the body's immunesystem attacks and destroys insulin-producing beta cells in the pancreas. As a result, the body produces little to no insulin, causing glucose to build up in the blood instead of entering cells for energy. Though the exact cause is not fully understood, it is believed to involve genetic and environmental factors. Symptoms often appear suddenly excessive thirst, frequent urination, fatigue, weight loss, and blurred vision and without proper care, it can lead to serious long-term complications.

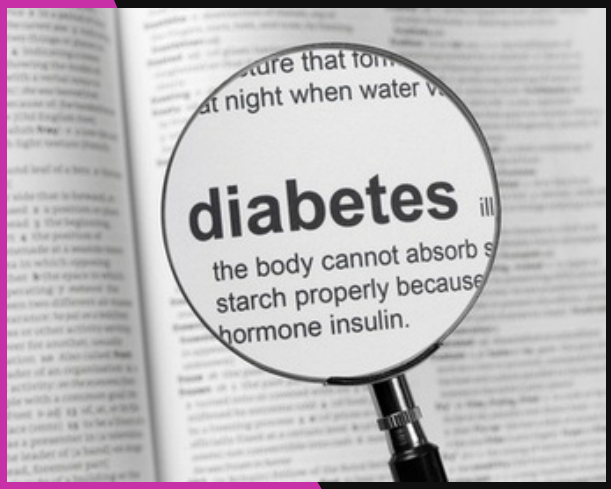
Managing this condition requires daily insulin and continuous monitoring of glucose levels, and this is where Continuous Glucose Monitoring (CGM) plays a transformative role. Traditionally, people rely on finger- prick glucometers, which require checking blood sugar multiple times a day often before and after meals, sometimes up to 6-7 times daily. This process can be painful and uncomfortable, especially for children , and still provides only



Apsana A

Student, University College of Engineering, Kariavattom

single, momentary readings. In contrast, a CGM system uses a small sensor inserted just under the skin, usually on the abdomen or arm, to measure glucose level in the interstitial fluid. The sensor contains an enzyme (glucose oxidase) that reacts with glucose and generates electrical signals, which is converted into real-time readings.







These readings are sent automatically to a smartphone or receiver every few minutes, allowing continuous tracking throughout the day and night. CGM not only shows current levels but also trends whether glucose is rising, falling, or stable-and provides alerts for highs and lows, helping prevent emergencies. It also introduces the concept of "Time in Range" giving a clearer picture of overall control. However, CGM comes at a cost. typically ranging from 3,000 to 6,000 per sensor in India, with each sensor lasting about 10 to 14 days, making it a recurring expense. Despite this, its ability to reduce pain, provide constant insights, and support better decision- making makes it a powerful tool in improving diabetes management and quality of life.



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